

Roll number:2403A51457

AI Assisted Coding Assignment(8.1)

TASK-1:

The screenshot displays the Visual Studio Code interface with the AI-assisted coding extension. The Explorer panel on the left shows a project named 'AI ASSISTED CODING' with five tasks listed: TASK-1, TASK-2, TASK-3, TASK-4, and TASK-5. The main editor window is open to 'TASK-1', showing a Python function `is_strong_password` and its test cases. The function checks for password length (at least 8 characters), presence of spaces, uppercase letters, lowercase letters, digits, and special characters. The test cases include assertions for valid and invalid passwords. The Terminal panel at the bottom shows the execution of the code, with prompts for password input and corresponding boolean outputs. The Chat panel on the right provides a natural language prompt for generating the code and test cases, along with a reference to the code in the editor.

```
1 import re
2
3 def is_strong_password(password):
4     # Check length
5     if len(password) < 8:
6         return False
7     # Check for spaces
8     if ' ' in password:
9         return False
10    # Check for uppercase
11    if not re.search(r'[A-Z]', password):
12        return False
13    # Check for lowercase
14    if not re.search(r'[a-z]', password):
15        return False
16    # Check for digit
17    if not re.search(r'[0-9]', password):
18        return False
19    # Check for special character
20    if not re.search(r'[!@#$%^&*~]', password):
21        return False
22    return True
23
24 # Assert test cases
25 assert is_strong_password("Abcd@123") == True
```

PS C:\Users\DELL\AI assisted coding> & "C:\Program Files\Python313\python.exe" "c:/Users/DELL/ai assisted coding/TASK-1"

Enter password 1: Abcd@123
Password: Abcd@123, Strong: True
Enter password 2: abcd123
Password: abcd123, Strong: False
Enter password 3: ABCD@123a
Password: ABCD@123a, Strong: True
PS C:\Users\DELL\AI assisted coding>

generate a python code to at least 3 assert test cases for is_strong_password(password) and implement the validator function.Password must have at least 8 characters.
o Must include uppercase, lowercase, digit, and special character.
o Must not contain spaces.Example Assert Test Cases:
assert is_strong_password("Abcd@123") == True
assert is_strong_password("abcd123") == False
assert is_strong_password("ABCD@1234") == True.Password validation logic passing all test cases.to check to enter and print those passwords

```
import re

def is_strong_password(password):
    # Check length
    if len(password) < 8:
        return False
    # Check for spaces
    if ' ' in password:
        return False
    # Check for uppercase
    if not re.search(r'[A-Z]', password):
        return False
```

Used 1 reference

Add Context... TASK-1 X

Add context (#), extensions (@), commands (/)

Ask GPT-4.1

TASK-2:



TASK-4:

The screenshot shows the VS Code interface with the following components:

- EXPLORER:** A sidebar on the left showing a file tree with `.vscode`, `# python.txt`, and five task files (`TASK-1` through `TASK-5`). `TASK-4` is selected.
- EDITOR:** The main workspace displays the code for `TASK-4`. It contains a Python class `Inventory` with methods `__init__`, `add_item`, `remove_item`, and `get_stock`. Below the class, there are assert-based tests for adding and removing items, and a `print` statement to show the stock after an addition.
- TERMINAL:** A terminal window at the bottom shows the command `PS C:\Users\DELL\AI assisted coding> & "C:\Program Files\Python313\python.exe" "c:/Users/DELL/ai assisted coding/TASK-4"` and its output:

```
Pen stock after adding 10: 10
Pen stock after removing 5: 5
Book stock after adding 3: 3
PS C:\Users\DELL\AI assisted coding>
```
- CHAT:** A chat window on the right provides a prompt to generate Python code with at least 3 assert-based tests for an `Inventory` class. It lists methods to implement and provides example test cases and expected output.

TASK-5:

